

LETLHAKANE, Botswana

WMO: 680400

Lat: 21.417S

Lon: 25.600E

Elev: 985

StdP: 90.04

Time Zone: 2.00 (E02)

Period: 96-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.0	6.5	-8.0	2.1	14.8	-5.9	2.6	14.4	8.5	17.9	7.8	18.1	2.1	160	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	13.1	36.9	18.9	35.9	19.3	35.0	19.3	23.0	29.4	22.5	29.0	22.1	28.7	3.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	18.1	24.7	21.0	17.7	24.3	20.5	17.1	24.2	73.6	29.4	71.8	29.0	70.1	28.7	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.3	6.4	DB	1.9	38.9	1.8	0.9	0.6	39.5	-0.4	40.0	-1.4	40.5	-2.7	41.2	(4)
(5)			WB	-1.4	24.6	1.5	0.7	-2.5	25.1	-3.4	25.5	-4.2	25.9	-5.3	26.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.6	26.3	26.0	24.7	22.4	18.9	16.6	15.9	19.2	22.9	25.9	26.4	26.2	(6)	
	DBStd	4.47	2.21	1.82	2.26	2.20	2.42	2.23	1.95	2.76	2.67	2.44	2.59	2.07	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	186	0	0	1	2	22	60	78	22	2	0	0	0	(9)	
	CDD10.0	4599	506	448	456	372	275	199	183	284	386	493	493	503	(10)	
	CDD18.3	1744	248	215	198	123	39	9	3	48	138	235	243	245	(11)	
	CDH23.3	19901	2563	2118	1902	1316	701	233	218	804	1772	2885	2854	2536	(12)	
(13)	CDH26.7	9304	1241	992	852	510	185	34	28	258	878	1582	1512	1232	(13)	
(14)	Wind	WSAvg	2.7	2.7	2.6	2.5	2.5	2.5	2.5	2.9	3.3	3.6	3.1	2.8	(14)	
(15) Precipitation	PrecAvg	386	84	79	58	12	5	2	0	0	3	17	51	69	(15)	
	PrecMax	816	242	271	166	31	39	34	2	2	18	43	104	108	(16)	
	PrecMin	133	2	2	1	7	0	0	0	0	0	4	3	28	(17)	
	PrecStd	177	66	72	58	6	9	8	0	0	6	11	33	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	36.6	36.1	34.8	32.3	29.2	29.8	33.0	36.8	37.9	38.2	36.7	(19)	
		MCWB	20.5	20.7	20.1	20.0	16.6	14.9	14.2	16.1	17.7	18.2	19.0	19.6	(20)	
	2%	DB	35.7	35.0	34.8	32.2	30.2	27.6	27.3	31.0	34.8	36.8	36.5	35.2	(21)	
		MCWB	20.4	20.3	20.0	18.2	16.1	15.0	13.5	15.5	17.0	17.9	19.1	20.0	(22)	
	5%	DB	34.1	33.5	33.1	31.1	28.9	26.0	25.9	29.4	33.2	35.2	35.1	34.1	(23)	
		MCWB	20.3	20.5	19.7	17.9	15.8	14.4	13.0	15.3	16.5	18.2	19.1	20.1	(24)	
	10%	DB	32.8	32.2	31.2	29.9	27.2	24.4	24.1	27.9	31.9	34.0	33.8	32.9	(25)	
		MCWB	20.4	20.5	19.7	17.9	15.3	13.7	12.9	14.7	16.4	18.1	19.1	20.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.6	23.5	23.0	22.1	19.1	18.0	16.5	18.5	20.5	21.7	22.4	23.4	(27)	
		MCDB	29.4	29.4	29.5	29.9	25.8	25.5	26.1	28.6	31.2	32.6	31.2	30.3	(28)	
	2%	WB	22.9	22.8	22.2	21.0	17.7	16.6	14.9	17.1	19.1	20.7	21.7	22.6	(29)	
		MCDB	28.5	29.4	28.6	28.7	27.2	24.1	24.7	28.4	30.6	31.6	30.3	29.5	(30)	
	5%	WB	22.4	22.4	21.7	20.0	16.9	15.6	14.1	16.1	18.1	20.0	21.1	22.1	(31)	
		MCDB	28.2	28.9	28.1	26.8	26.8	23.2	23.4	27.3	29.8	31.1	29.9	29.1	(32)	
	10%	WB	21.9	21.9	21.2	19.1	16.1	14.5	13.3	15.3	17.3	19.3	20.6	21.6	(33)	
		MCDB	27.9	28.2	27.8	26.0	25.6	22.3	22.7	26.3	28.9	29.9	29.1	28.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	11.8	12.0	14.3	16.7	15.5	16.5	16.7	16.5	14.6	13.1	11.9	(35)	
		MCDBR	14.3	14.4	15.0	16.2	18.4	17.9	19.5	18.4	18.5	16.9	15.4	14.2	(36)	
		MCWBR	4.8	4.7	5.3	6.4	8.2	8.8	9.8	8.6	8.1	6.5	5.3	4.4	(37)	
	5% WB	MCDBR	10.9	10.9	11.1	13.1	16.1	14.8	16.4	17.0	16.1	14.2	12.6	11.9	(38)	
		MCWBR	4.1	4.0	4.3	5.5	7.4	7.7	8.7	8.3	7.4	5.6	4.5	4.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.372	0.373	0.360	0.332	0.297	0.289	0.290	0.355	0.426	0.429	0.386	0.376	(40)	
	taud		2.417	2.409	2.437	2.460	2.505	2.494	2.481	2.277	2.098	2.149	2.318	2.394	(41)	
	Ebn at Noon		972	958	946	931	930	918	929	893	866	895	952	968	(42)	
	Edh at Noon		126	125	116	105	92	89	93	124	160	160	138	129	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.04	6.68	6.25	5.68	5.15	4.57	4.86	5.65	6.53	7.02	7.02	6.86	(44)	
(45)	RadStd		0.48	0.61	0.47	0.39	0.14	0.20	0.13	0.25	0.30	0.35	0.47	0.45	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page